Plumber's guide: the studio

Everything the plumber needs for one container studio: supply, hot water, drains, the tests, and what the studio needs from the farm.

Status: revision A2, 2026-10-28, PRELIMINARY. No guest stays before the septic system passes its inspection (D-016).

The two studios are piped the same, so one guide serves both.

Kinyarwanda: soma aya mabwiriza mu Kinyarwanda · Print: this guide as a PDF

Words used here

- **End A**: the closed end wall, by the bed. **End B**: the cargo-door end, by the bathroom.
- **Back wall**: the long wall with no openings. **Front wall**: the long wall with the sliding glass door.
- Positions in the fittings table are millimetres from the finished end-A wall (x), from the finished back wall (y) and up from the finished floor (z).

The studio at a glance

What	Design
Supply pipe	PPR PN20, 25 mm for the main, 20 mm inside
Main in	under the floor at end B, up into the sink base: stop valve, Y-filter, check meter (SV)
Hot water	a 30 L box, 2 kW, IPX4, on end wall B above the WC; stored at 60 °C
Mixing valve (TMV)	behind a tiled access panel in end wall B, between the shower and the WC; 41 to 44 °C to the shower and the basin
Drains	WC 110 mm, shower 50 mm, basin 40 mm, kitchen sink 50 mm on its own line to a grease trap; all at 1:40 or steeper
Vent	75 mm open vent up end B, ending 900 mm above the roof guardrail
Shower tray	1000 × 900
Pressure	2.5 to 3 bar from the farm's booster pump

When	What	Step
After the battens (S03)	First fix: main in, every supply and drain run, the 10 bar test	S04
With the bathroom	Tray (S09), WC, heater and basin (S10)	S09, S10
With the kitchen	Sink, tap and filter	S13

Three hold points

1. **Battens first.** Start the first fix only when the battens (S03) are fixed and the floor sheets are down.
2. **The 10 bar test.** Fill the PPR, hold 10 bar for one hour with no drop, write it down and sign it before the spray foam (S05). The foam buries the pipes.
3. **Before guests.** The drains are fill-tested, the vent smoke-tested, the TMV measured at the shower, and the farm's septic system inspected.

S04: first fix

1. Bring the cold main (W1, 25 mm) in under the floor at end B and up into the sink base through the stop valve, Y-filter and check meter, where it can be shut off without tools.
2. Run the cold inside the units and the walls: through the partition, the back wall's lining and end wall B to the WC cistern (W4), the basin (W5) and up to the heater (W2). Keep cold 300 mm above the floor.
3. Run the hot (60 °C) down from the heater to the mixing valve (H1) and back along the same walls to the sink (H2, labelled hot). Keep hot 360 mm above the floor.
4. From the mixing valve, take the two tempered runs (T1 to the shower valve, T2 to the basin): each about 1.2 m, under the 2 m allowed for a mixed-water dead leg.
5. Every run goes straight: along the wall, then straight up to its outlet. No diagonal legs.
6. Hang the drains in the void under the floor (the container stands about 700 mm up): WC 110 mm, shower 50 mm, basin 40 mm to the inspection chamber; the kitchen sink 50 mm on its own line to the grease trap. Pass between the 122 mm cross-members, never through them.
7. Fall at 1:40 or steeper; rodding eyes at every bend; the 75 mm vent up end B.
8. Pressure-test at 10 bar for one hour, photograph every wall, sign the release for the foam.

Pipe runs: system, size, length and fall · download CSV

run	system	dn	length_m	fall	note	colour
D1	soil	110	1.8	1:38	WC to IC1, 1:40	#8b4a12
D2	waste	50	1.3	1:34	Shower (90 mm tray waste) joins the soil run	#8b4a12
D3	waste	40	2.1	1:28	Basin (wash lobby) joins the soil run	#8b4a12
D4	waste	50	3.3	1:26	Kitchen sink to the grease trap (separate soak pit)	#8b4a12
V1	vent	75	6.7		Open vent up end B to 900 above the guardrail	#9a9893
W1	cold	25	2.9		Cold main in from the farm, up into the sink base: stop valve, filter, check meter	#2a78d6
W2	cold	20	1.8		Cold up end wall B to the heater	#2a78d6
W3	cold	20	0.1		Cold to the sink's angle valve	#2a78d6
W4	cold	20	5		Cold through the partition and the walls to the WC cistern	#2a78d6
W5	cold	20	1		Cold on to the basin	#2a78d6
W6	cold	20	0.1		Cold to the mixing valve	#2a78d6
H1	hot	20	2.4		Hot (60 C) from the heater down to the mixing valve	#e34948
H2	hot	20	3.9		Hot back along the walls to the sink (direct 60 C, labelled)	#e34948
T1	tempered	20	1.2		TMV 41-44 C up to the shower valve	#eda100
T2	tempered	20	1.2		Tempered to basin	#eda100

Fittings: what, where and how high · download CSV

point	kind	model	runs	description	item	x_mm	y_mm
SV	valve	meter	W1	Stop valve, Y-filter and check meter on the main, in the sink base	Stop valve + Y-filter + check water meter	4578	1540
WH	heater	heater	W2 H1 H2	30 L water heater, a box on end wall B above the WC, its T&P valve piped outside	Electric storage water heater 30 L, 2 kW, IPX4, with T&P valve	5768	1300
TMV	valve	tmv	H1 W6 T1 T2	Mixing valve behind an access panel in end wall B, between the shower and the WC	Thermostatic mixing valve (TMV) 41-44 C	5768	1005
KT	tap	kitchentap	W3 H2 D4	Kitchen mixer tap, angle valves under the sink	Kitchen mixer tap	4528	1375
DF	filter	filter	W3	Drinking-water filter in the sink base, its own faucet by the tap	Under-sink drinking-water filter + faucet	4278	1168
SH	shower	showerset	T1 D2	Thermostatic shower valve, head and rail on end wall B	Thermostatic shower valve + 6-7 L/min head + rail	5768	450

point	kind	model	runs	description	item	x_mm	y_mm
WC	wc	cistern	W4 D1	Concealed cistern in end wall B, dual-flush plate	Slim concealed cistern frame (80 mm) with dual-flush plate, in the end-B stud wall and its 30 mm build-out	5768	1300
BT	tap	basintap	W5 T2 D3	Basin mixer, angle valves under the console	Wall basin 400 x 300 + mixer + bottle trap (wash lobby)	5768	2020
IC1	site	chamber	D1 D2 D3	Inspection chamber outside end B, on to the farm's septic tank	Septic tank (2 chambers, about 4 m3) + inspection chambers	6698	1126
GT	site	greasetrap	D4	To the farm's grease trap and its own soak pit	Grease trap + kitchen soak pit	6698	1375
VT	vent	vent	V1	Vent terminal, 900 mm above the roof guardrail	uPVC vent pipe DN75	5918	200

S09, S10 and S13: second fix

1. S09: set the shower tray (1000 × 900) on its 90 mm waste, with the carpenter's ledge and the fixed glass screen.
2. S10: hang the wall-hung WC on its slim concealed cistern (80 mm frame in end wall B, 30 mm build-out, dual-flush plate).
3. S10: hang the 30 L heater box (450 × 450 × 300) above the WC; pipe its temperature-and-pressure relief valve outside.

4. S10: fit the basin (400 × 300) and its mixer with a bottle trap in the wash lobby; the thermostatic shower valve, head and rail on end wall B.
5. S13: the kitchen mixer tap on angle valves, and the drinking-water filter in the sink base with its own faucet by the tap.

Tests

1. PPR at 10 bar for one hour, before the foam (S04).
2. Fill-test every drain and check every trap holds its seal.
3. Smoke-test the vent.
4. Measure the mixed water at the shower with a thermometer: 41 to 44 °C.
5. Run a hot shower, then time the heater: about 40 minutes to heat up again.

What the studio needs from the farm

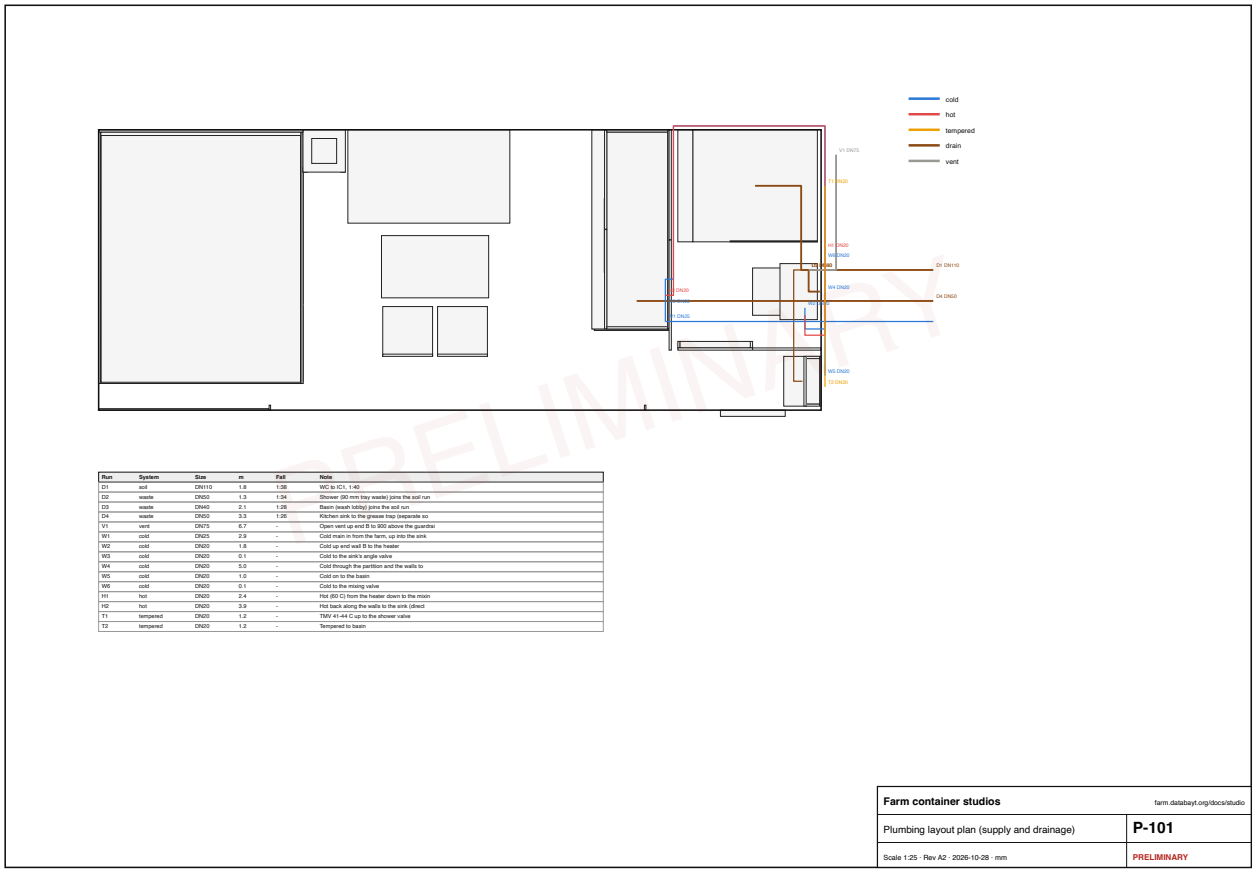
- One cold supply at 2.5 to 3 bar from the farm's storage tank and booster pump. Mains water enters that tank through an air gap, so private water never flows back into the main (D-016).
- A soil connection from the inspection chamber outside end B to the farm's septic tank, and the kitchen line to a grease trap with its own soak pit.
- The septic tank sits at least 50 m from the borehole, 3 to 7 m from buildings and the boundary, downhill of the borehole, within 30 m of tanker access, and only after a percolation test.

Site water and septic sizing for the two studios · download CSV

item	value
persons	4
daily_L	600
tank_m3	2.5
septic_rs143_m3	3
septic_2012_m3	2.72
septic_uk_m3	3.08
septic_m3	3.5
chamber1_m3	2.33
chamber2_m3	1.17

item	value
pump_lpm	26

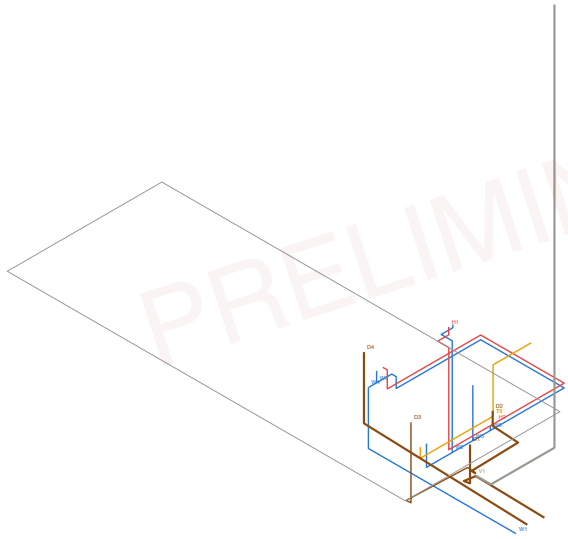
Drawings



Run	System	Size	m	Fall	Note
D1	soil	DN110	1.8	1:38	WC to IC1, 1:40
D2	waste	DN50	1.3	1:34	Shower (80 mm tray waste) joins the soil run
D3	waste	DN40	2.1	1:28	Basin (wash lobby) joins the soil run
D4	waste	DN50	3.3	1:26	Kitchen sink to the grease trap (separate so
V1	vent	DN75	6.7	-	Open vent up and B to 900 above the guardroi
W1	cold	DN25	2.9	-	Cold main in from the farm, up into the sink
W2	cold	DN20	1.8	-	Cold up and wall B to the heater
W3	cold	DN20	0.1	-	Cold to the sink's angle valve
W4	cold	DN20	5.0	-	Cold through the partition and the walls to
W5	cold	DN20	1.0	-	Cold on to the basin
W6	cold	DN20	0.1	-	Cold to the mixing valve
H1	hot	DN20	2.4	-	Hot (60 C) from the heater down to the mixin
H2	hot	DN20	3.9	-	Hot back along the walls to the sink (direct
T1	tempered	DN20	1.2	-	TMV 41-44 C up to the shower valve
T2	tempered	DN20	1.2	-	Tempered to basin

P-101 Plumbing layout plan · open full size

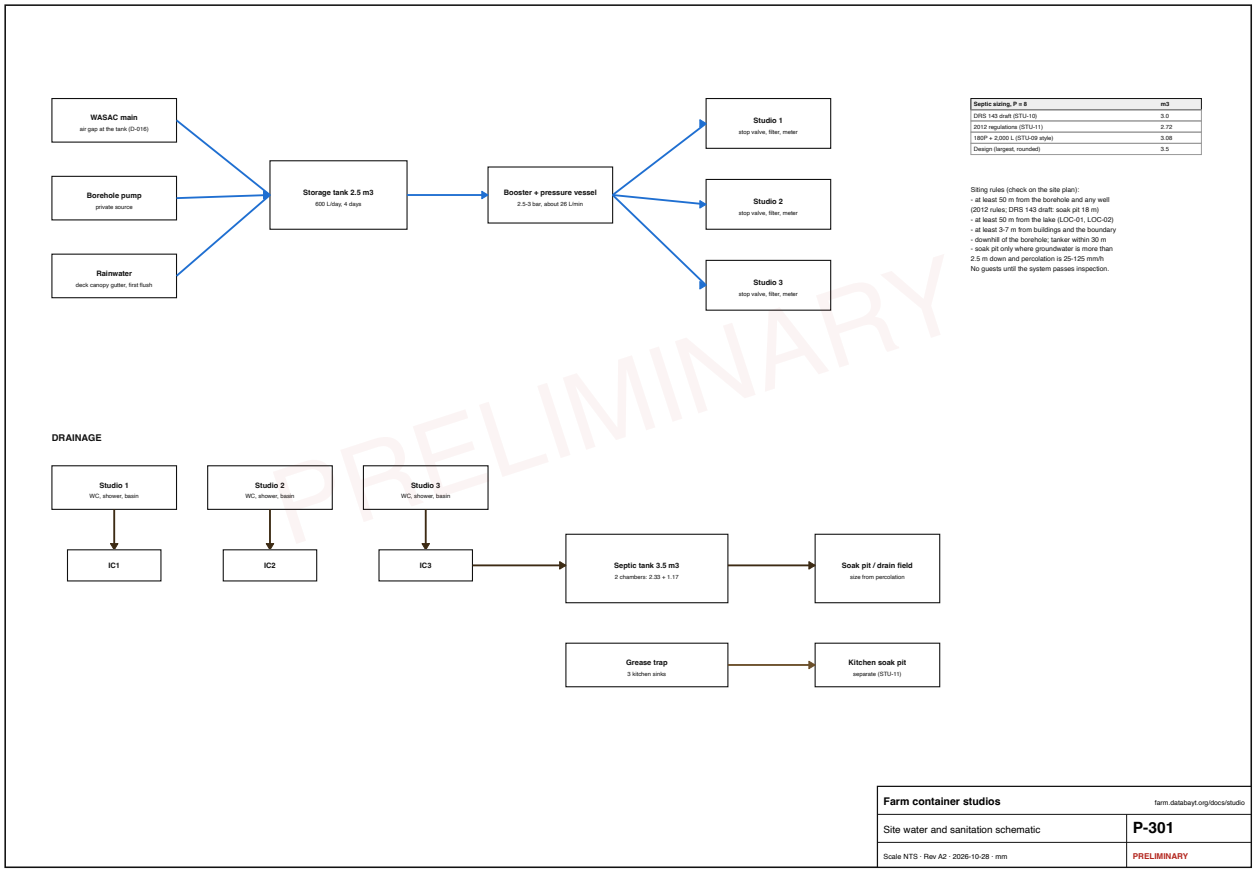
Supply: PPR PN20, 20/25 mm. Hot leaves the 30 L heater at 60 C; a thermostatic mixer (TMV) delivers 41-44 C to the shower and basin; the kitchen sink gets direct hot (labelled).
Drainage hangs in the void under the raised floor (the container is about 700 mm up), between cross-members: WC 110, shower and sink 50, basin 40, falls at least 1:40.
Vent: 75 mm open vent up end B to 900 above the guardrail line (or an AAV inside).
Kitchen sink waste runs separately to a grease trap and its own soak pit (STU-11).



PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Pipework isometric (unit)	**P-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

P-201 Pipework isometric · open full size



P-301 Site water and sanitation schematic · open full size